Mohamed Mostafa Nazih

Backend and Platform Software Engineer

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Professional Experience

Oblien **Aug 2025 – Present**

Founding Engineer / Part-time, remote

- Designed and maintained **Openship**'s TypeScript/Hono control plane, giving API, SDK, CLI, and desktop clients shared authorization and deployment behavior across local, SSH, and cloud targets.
- Engineered deployment coordination and recovery using PostgreSQL advisory locks, durable execution receipts, and release snapshots to serialize host operations and recover interrupted builds without reclaiming live workers' resources. Added release gates with Docker integration tests.
- Built **Mindwire**'s Go runtime and TypeScript SDK for local, SSH, and Docker agents; persisted turn receipts before execution to deduplicate retries and reconciled unfinished runs after daemon restarts.
- Developed **BytesMap**'s query/reporting backend with connection leases, row/byte/time limits, encrypted credentials, and transactional report claims. Built Go ingestion that acknowledges events after durable JetStream publication and indexes them in ClickHouse.
- Built Firecracker VM lifecycle and network policies; implemented billing-aware capacity recovery with quota rechecks and stop tokens to avoid restarting manually stopped workspaces.
- Developed **Reflx**'s Express/Prisma backend from profile ingestion through agent conversations and WhatsApp integration, retaining guest/member session state and conversation history across requests.

Medica Scope **Jul 2025 – Present**

Backend Engineer / Full-time, hybrid

- Owned **CarLink**'s NestJS/Prisma/PostgreSQL backend implementation across authentication, vehicle integrations, bookings, and workshop operations. Used conditional status updates to enforce booking transitions and retained quoted prices for invoicing.
- Implemented CarLink consent/deletion workflows, append-only audit ledgers, and retention controls to keep administrative and privacy actions traceable; maintained migrations and PostgreSQL-backed CI tests.
- Developed **Rewaaq**'s .NET 10 course/quiz APIs with subscription-scoped access, persisted question selection, grading, and server-enforced attempt deadlines. Built GitLab deployment pipelines with health checks and rollback for authentication and application services.
- Built **CareLinker** patient/case and messaging subsystems with record-level authorization, transactions, and encrypted uploads; persisted conversation history and rechecked case access for REST/Socket.IO messaging.
- Connected **Medica Academy**'s Paymob checkout/webhooks to course enrollment and VdoCipher playback, with transactional enrollment creation and lesson-access checks before issuing video credentials.

Onvo **May 2024 – Jul 2025**

Software Engineer / Full-time, remote

- Reworked ONVO's PHP backend in Express.js, including anonymous Q&A, threaded replies, and followed-user feeds with capped queries and timestamp-based pagination.
- Implemented messaging permissions, replies/media, and push notifications; used Redis session/context lookups and rate limits, request-size limits, and XSS middleware around backend endpoints.
- Built Airwave's Node.js backend for music/podcast playback, with playlists and WebSocket listening rooms that synchronize playback state across connected clients.

Selected Projects

LLMProxy github.com/Mo7ammedd/LLMProxy

C#, .NET 10, PostgreSQL, Redis | Personal project

- Designed a gateway with 10 provider adapters, atomic Redis rate limits, and SQL token/spend reservations to enforce shared allowances under concurrent requests.
- Made usage finalization idempotent and recovered abandoned reservations conservatively; limited failover to before the first stream event to avoid replaying partial responses. Added PostgreSQL/Redis integration tests and CI recovery drills.

GhostyKey github.com/Mo7ammedd/GhostyKey

TypeScript, Next.js, PostgreSQL | Personal project

- Built browser-encrypted secret sharing with AES-GCM and fragment-only decryption keys; the server stores ciphertext and hashed read/deletion capabilities.
- Serialized consumption, expiry checks, and deletion in PostgreSQL transactions. Verified single-use consumption with 20 concurrent readers and expiry after lock waits in database tests.

AeroUDP github.com/Mo7ammedd/AeroUDP

Rust, Tokio | Experimental transport protocol

- Implemented experimental ordered UDP transport with sequence-numbered packets, acknowledgements, sliding windows, retransmission, and out-of-order buffering.
- Added TCP-inspired congestion recovery, RTT-based timeouts, and separate receiver/congestion windows; verified codec properties, connection handshakes, and loopback transfers with Rust tests.

Technical Skills

Languages: C#, TypeScript, Go, JavaScript; Rust and Python (project work).

Backend: ASP.NET Core/.NET, NestJS, Hono, Express.js, Node.js, Bun.

Databases and caching: PostgreSQL, MySQL/MariaDB, SQL Server, SQLite, Redis, ClickHouse.

Data access: Entity Framework Core, Prisma, Drizzle.

Infrastructure / CI/CD: Linux, Docker, Firecracker, Nginx/OpenResty, SSH; GitHub Actions, GitLab CI.

APIs and messaging: REST, SSE, Socket.IO, JWT/OTP, NATS JetStream, RabbitMQ.

Testing and tools: xUnit, Bun test, Go testing, Docker E2E, OpenAPI, Git.

Education

Suez Canal University **Oct 2021 – Jul 2025**

B.Sc. in Computer Science. Graduation project grade: Excellent.
Mentored students in backend development through Google Developer Groups.